



Article

# Predicting Trends and Maximizing Sales: AI's Role in Saudi E-Commerce Decision-Making

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## Abstract

Artificial intelligence (AI) has emerged as a transformative force across various sectors, providing innovative solutions and enhancing operational processes. In the e-commerce domain, AI has significantly contributed to customer-centric approaches and strategic decision-making, fostering superior customer experiences. This study investigates the role and impact of AI in the Saudi e-commerce sector, focusing on the perspectives of female customers and retailers. Grounded in sociotechnical theory, the research employs a mixed-methods approach, combining quantitative surveys and semi-structured interviews. The quantitative findings demonstrate that AI-enabled e-commerce positively influences customer experience, customer satisfaction, and operational efficiency. Key AI capabilities, such as task automation, personalized recommendations, and predictive analytics, enhance online retail systems' performance. The qualitative analysis highlights both the opportunities and challenges associated with AI adoption, emphasizing the need for specialized infrastructure and skilled professionals. Participants recommend addressing the skill gap and adopting phased implementation strategies to optimize AI integration. This study provides actionable insights and strategic recommendations for policymakers and stakeholders in the Saudi e-commerce sector.

**Keywords:** artificial intelligence; e-commerce; customer satisfaction; operations optimization; Saudi Arabia; sociotechnical theory



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## 1. Introduction

The e-commerce sector in Saudi Arabia is experiencing rapid growth due to successive technological developments and changing consumer behavior, especially with the increased reliance on the internet and digital applications in shopping processes [1]. In this context, the use of artificial intelligence (AI) has become an essential tool for supporting business decision-making and analyzing big data to predict future trends, improve customer experience, and maximize sales [2]. AI also enables companies to understand consumer behavioral patterns, forecast demand, improve inventory management, and customize marketing offers to suit individual customer preferences [3]. AI applications are expanding equally, facilitating commercial arenas AI can gather, analyze [4], and interpret vast amounts of data, allowing for it to make proactive decisions specifically about AI for commercial purposes, electronic commerce, also known as e-commerce, is one of the significant domains where the relevant technology is widely adopted and implemented [5]. Today, several e-commerce companies have adopted different types of AI-enabled smart systems to better understand their customers, demands, interests, and experiences. Since its inception more than 60 years ago, AI has significantly contributed to the developments

now integral to social and economic well-being and development. Specifically regarding AI in e-commerce, this field has experienced a remarkable technological transformation in recent years [6]. The convenience offered by e-commerce continues to grow in popularity, with customer expectations continuously rising. In this context, AI is critical in driving e-commerce growth by introducing innovative trends and ideas [7]. Thus, the rise of e-commerce marketing in Saudi Arabia presents significant growth opportunities for the country's retail sector [8]. This expansion promotes an attractive environment for both local and international retail companies. Improving national connectivity infrastructure and expanding broadband reach directly contribute to the e-commerce industry's growth [2]. Consequently, today, Saudi Arabia is at the forefront of AI technology adoption in e-commerce due to its cognitive methods of processing and interaction. As Saudi Arabia diversifies from its traditional oil reliance, startups obtain support to penetrate new markets. AI enables these companies to target the right customer segments and focus their marketing strategies effectively, providing the potential to upgrade their operations. AI and entrepreneurs can better serve customer needs through changes in culture, policies, practices, business regulations, and transparency in big data collection [9,10]. According to a report by Internet World Usage and Population Statistics, around 36.84 million people in Saudi Arabia have access to the Internet [11]. However, this internet indicates a robust increase in daily users during and after the COVID-19 pandemic [12], indicating a greater reliance on e-commerce among Saudi consumers. Considering the importance of AI in e-commerce, this research examines the quantitative and qualitative aspects of the Saudi e-commerce industry [7].

Artificial intelligence (AI) technologies enable companies to identify customer behavior patterns, predict future demand, manage inventory more efficiently, and customize marketing offers to match individual customer needs and preferences [13,14]. Furthermore, AI contributes to improving companies' operational performance by automating complex business processes, reducing waste, and enhancing the effectiveness of digital marketing campaigns [15]. Despite the significant benefits of AI technologies, their implementation in the Saudi market still faces challenges, such as limited technical expertise among some organizations, data protection issues, and the need to integrate smart analytics with traditional marketing strategies [16,17]. Therefore, this research aims to examine how AI can be used to support e-commerce decision-making in Saudi Arabia, focusing on its role in predicting trends and maximizing market opportunities.

**Thus, this research addresses the following three research questions:**

**RQ1.** How does AI affect female customer experiences in the Saudi Arabian e-commerce industry?

**RQ2.** How does AI affect female customer satisfaction in the Saudi Arabian e-commerce industry?

**RQ3.** How does AI affect e-commerce operations in the Saudi Arabian e-commerce industry?

## 2. Literature Review

E-commerce and information management have witnessed remarkable development globally over the past two decades, with digital transformation becoming a key focus for improving corporate performance and enhancing customer experience [18–20]. In the European context, studies have shown that the integration of e-commerce with information management systems leads to improved demand forecasting accuracy and increased sales, with a focus on the use of advanced analytics and artificial intelligence to support decision-making [21,22]. In Asia, research in China and Japan has highlighted the importance of big data and artificial intelligence in improving logistics operations, enhancing consumer

digital trust, and providing secure and efficient electronic payment solutions [23,24]. In the Gulf Cooperation Council (GCC) region, recent studies have indicated that e-commerce is experiencing rapid growth driven by government digital transformation initiatives, such as those in the UAE, Saudi Arabia, and Kuwait, with a focus on developing digital infrastructure and improving data management to enhance user experience and raise operational efficiency [25]. Studies have also shown that integration between e-commerce platforms and information management systems contributes to enhancing customer satisfaction and loyalty, highlighting the importance of effective use of data in formulating accurate marketing strategies [26–29]. In the Saudi context, local research has focused on the development of e-commerce in recent years in light of Saudi Vision 2030, which emphasizes the digital economy, smart government transformation, and financial services [8]. Studies have shown that the success of e-commerce in the Kingdom depends largely on information management, cybersecurity, and the provision of reliable electronic payment methods [30,31]. Research has also highlighted the importance of understanding Saudi consumer behavior and digital preferences to design effective user experiences [32], as well as the need to adopt advanced data strategies to support decision-making and improve the overall performance of e-platforms [33].

### 2.1. Sociotechnical Theory

This research is theoretically supported by sociotechnical theory [34]. It has the proposition that the design, implementation, and performance of any organization can only be comprehended and enhanced if both technical and social factors are brought together and treated as interdependent aspects of a system [35,36]. Accordingly, [31] argued that sustaining an organizational operation and integrating technologies is challenging for stakeholders [30]. However, today, organizations are influenced by technological developments and need more customer-centric systems and approaches to help acquire organizational objectives and meet customer expectations. Notably, [36] considers the integration of technologies in organizational operations as a complex task due to the different areas and systems involved, and it is sensitive because of the complicated relationship among these systems within the environment [37]. Any alterations aiming at supporting organizational development are technically involved and equally considered regarding how these interdependent relationships would impact the organizational environment and operations [38]. Therefore, this research highlights the implementation and use of AI in the Saudi e-commerce industry due to customer experiences, satisfaction, and the technical prospects offered by the technology. Considering these benefits, it is assumed that e-commerce stakeholders in Saudi Arabia are actively adopting more customer-centric approaches that focus on their financial gains and fulfilling customer demands in the most effective possible manner [39]. From this perspective, how smart technologies such as demand forecasting systems and artificial intelligence impact customer decisions and organizational processes is analyzed, taking into account the social interactions between users and employees. The concepts of “joint optimization” and “interconnectedness between systems” are emphasized to link theory to practical results. The social-technical theory provides a holistic approach to examine the responses of Saudi female e-commerce customers, reflecting their individual experiences and perceptions [40]. In addition, it also helps to explore how customer experiences affect the decision-making among e-commerce companies, which further helped them to design, develop, and implement artificial intelligence-enabled technologies in their systems [41].

## 2.2. AI and Customer Experience in E-Commerce

The advent of the internet and technological advancement has changed how companies work and customers shop, driving the quick expansion of the e-commerce industry. E-commerce platforms have become ubiquitous, providing customers with exceptional convenience, variety, and access to services and products [42]. As online shopping acquires global momentum, e-commerce companies aim to meet customers' evolving expectations in a competitive landscape. Within this fast-changing environment, artificial intelligence stands out as a disruptive force reshaping the e-commerce sectors [25]. AI technologies can analyze vast amounts of data, automate processes, and derive insights using machine learning, predictive analytics, and natural language processing [43]. To further affirm the role and impact of artificial intelligence in the e-commerce industry, Daqar and Smoudy [44] examined its impact on e-commerce customers in Palestine across different industries, including telecommunication and banking [45]. A mixed-methods approach revealed a significantly positive relationship between AI and customer experiences. On the other hand, customer experiences were analyzed in two dimensions, including customer services and after-sales support, indicating a significant and positive relationship between these variables [44]. Specifically, this study highlights that providing personalized customer services to a customer's buying journey significantly improved customer experience, leading to strong customer loyalty [46,47].

## 2.3. AI and Customer Satisfaction

AI-empowered recommendation systems are critical in improving customer experience and driving sales in e-commerce platforms. Existing research [48,49] also highlights the significance of personalized recommendations in boosting customer engagement and satisfaction [24]. By employing machine learning algorithms to analyse customer preferences and behavior, e-commerce companies can provide predesigned product suggestions, leading to higher conversion rates and customer retention. According to [50], AI in the e-commerce sector has introduced technology like chat boards and virtual assistants, which are transforming customer services in the e-commerce industry [49]. Research by [51], also highlighted the effectiveness of chat ports in providing personalized assistance, facilitating seamless transactions and answering customers' queries [50]. Accordingly, AI-powered chatbots with natural language processing capabilities can stimulate human-like interactions, promote improved consumer engagement, and facilitate shopping experiences [52]. This leads to improved customer loyalty due to satisfaction and improved shopping experiences [53]. Metsai et al. [49] examined the reasons behind brands success in e-commerce through the advancement of artificial intelligence through the lens of customer satisfaction by proactively meeting their needs without needing them to search for products and services actively. Data gathered from 50 online customers in India revealed that online shopping is heavily automated using AI, consistent with consumers' expectations and preferences [23]. As a result, the relevant technology has enhanced consumer and product brand associations, increasing customer loyalty and satisfaction in the Indian e-commerce sector [54].

## 2.4. AI in Optimizing E-Commerce Operations

Today, AI has improved the overall efficacy and performance of e-commerce systems worldwide. For example, the existing literature shows that predictive analytics techniques enabled by AI allow e-commerce companies to predict customer behavior, optimize inventory management, and predict market trends [55]. The study by [50] indicates that a predictive analytic model uses previous data to anticipate future purchase patterns, enabling retailers to make data-driven decisions concerning promotions, pricing, and product

allocation cost by accurately predicting demand, e-commerce companies can decline stock-outs, improve performance, and lower inventory costs. Another example of AI use in optimizing e-commerce operations can be cited from AI-driven supply chain optimization solutions that offer e-commerce companies prospects to improve logistics efficiency, fulfill orders, and reduce costs [15,28]. According to Müller (2022), AI technology, such as optimization algorithms and machine learning, provides real-time visibility into supply chain operations, facilitating risk management and proactive decision-making processes [22]. Today's e-commerce companies can streamline and improve their supply chain operations and reach customers' demands by optimizing inventory levels, warehouse management, and supply chain operations. Another important aspect can be establishing AI-based fraud detection systems crucial in securing e-commerce platforms against fraudulent activities and ensuring secure transactions. Research by [56] showed the effectiveness of AI algorithms in detecting anomalous patterns, identifying fraudulent transactions, and preventing potential risks [57]. AI-powered fraud detection system scans accurately differentiate between fraudulent and legitimate activities, improving credibility and trust among e-commerce customers. Thus, the reviewed literature helped us to hypothesize the following:

- H1.** *AI positively affects female Saudi customers' experience in the Saudi Arabian e-commerce industry.*
- H2.** *AI positively affects female Saudi customers' satisfaction in the Saudi Arabian e-commerce industry.*
- H3.** *AI has positively optimized e-commerce operations in the Saudi Arabian e-commerce industry.*

### 3. Methodology

This research comprises a mixed-methods design that helps gather qualitative and quantitative data and analysis [51,58–60]. Employing the mixed-methods approach in the current research helped provide in-depth insights about e-commerce from both consumer and company perspectives [61–63]. The current research's mixed-methods approach mainly involved quantitative surveys and semi-structured face-to-face interviews with the participants. Using the mixed-methods helped the triangulation process, which later involved merging and representing the results collaboratively for better insights [52,64–66]. Thus, using a mixed-methods approach facilitated in-depth results, and enables their organized and extensive representation and discussion in later sections [67].

#### 3.1. Study Population and Sampling Methods

The population of the current research involves female university-level students who prefer online shopping for purchasing products and online retailers of these products in Saudi Arabia. This selection intended to focus on a specific group whose behaviors and attitudes could be studied in-depth, in line with the research objectives [38]. However, to further narrow down the quantitative sample, two public sector higher-education institutions were selected, having a total population of 91,520 female students on different levels [39]. However, the precise sample size was calculated using Krejci and Morgan's calculation formula, indicating that a sample size of 382 respondents was suitable for the current study [68]. A simple random sampling method was employed for the selection of respondents, ensuring an equal chance of selection for each individual in the study [69,70]. The quantitative data was gathered from February 2024 to June 2024 through personal visits to the selected institutions after receiving formal permission from the selected institutions. Once the data was collected, the obtained questionnaires were calculated and evaluated [71–73]. This step indicated five questionnaires that needed to be completed by

the respondents. Thus, 377 questionnaires were finalized, indicating a response rate of 98.6%, higher than the minimum rate of 60% [71].

Furthermore, a sample size of seven participants was selected based on the suggestions by [74], who recommended a sample size of 5–25 participants as sufficient for this study. In addition, a convenience sampling approach was applied when selecting the participants [75]. The participants were contacted via email and were interviewed using online resources (Microsoft 365). The focus remained on accessing and interviewing the e-commerce business owners to understand their opinions and perspectives on AI implementation. Notably, informed consent was obtained from both the survey respondents and interview participants. Also, data confidentiality was ensured as a primary research ethic [76].

### 3.2. Data Collection Tools

Data was collected using structured surveys and semi-structured face-to-face interviews. Following the sequential mixed-method approach, surveys were conducted, followed by interviews. The survey tool was designed by adopting questionnaire items and scales from existing studies. These items were edited and reshaped to match the current study problem and objectives. All participants were informed of the study objectives and the nature of their participation, and their informed consent was obtained prior to any interviews or data collection. This study was also reviewed and approved by the Institutional Review Board/Ethics Committee at Princess Nourah bint Abdulrahman University review board of the academic institution to ensure full compliance with ethical standards. Table 1 represents the details of quantitative surveys for the data collection. Further, an interview protocol guide is designed for qualitative data gathering purposes. Table 2 represents the details of the interview protocol guide.

**Table 1.** Details of quantitative survey questionnaire.

| Variables                          | Questionnaire Items                                                                            | Sources |
|------------------------------------|------------------------------------------------------------------------------------------------|---------|
| Artificial Intelligence            | AI in online purchases is a positive technology addition today.                                | [77]    |
|                                    | AI in online purchases adds more value to our daily shopping experiences.                      |         |
|                                    | AI in online shopping and retail are positively affecting my product decision-making.          |         |
|                                    | Using AI features, i.e., chatbots and virtual assistants, is intuitive to buy online products. |         |
| Customer Experience                | I am provided with the right solutions by the AI bots used by the service providers.           | [78]    |
|                                    | My queries are adequately addressed after talking to the service provider.                     |         |
|                                    | I am engaged when AI chatbots collaboratively address my required products.                    |         |
|                                    | The visual cues on the service provider's website interface are convenient.                    |         |
| Customer Satisfaction              | I had a positive experience interacting with AI-enabled online retailers.                      | [79]    |
|                                    | I would readily recommend an online retail site that uses AI.                                  |         |
|                                    | My shopping experience with an AI-enabled online retail website has fulfilled my expectations. |         |
|                                    | I am generally satisfied with my purchases from an AI-enabled retailer.                        |         |
| E-commerce Operations Optimization | I believe that AI tools significantly improve the efficiency of online retail shop operations. | [80]    |
|                                    | AI helps optimize inventory management functions in online retail shops.                       |         |
|                                    | The use of AI contributes to more rapid order fulfillment in online purchase systems.          |         |
|                                    | I believe that today, online purchase decision-making is largely simplified by AI technology.  |         |

**Table 2.** Qualitative Interview Guide.

| Key Topic                                | Root Questions                                                                                                |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Effects of AI on Customers in E-Commerce | How do you consider AI implementation in the Saudi e-commerce sector?                                         |
|                                          | In your opinion, how does AI affect customers in Saudi online retail?                                         |
| Impact of AI on Overall Performance      | How would you describe the current framework of AI in Saudi e-commerce?                                       |
| Current Framework of AI in E-Commerce    | What recommendations would you suggest for improving the implementation of AI in the Saudi e-commerce sector? |

### 3.3. Data Analysis

This research involves quantitative and qualitative methods, so the analysis involved the relevant approaches. First, the quantitative analysis involved descriptive and inferential statistics, particularly the Partial Least Squares-Structural equation (PLS-SEM). Furthermore, thematic analysis was applied for the qualitative data analysis. In this regard, this study involves a sequential mixed-methods approach followed by quantitative and qualitative analyses.

## 4. Study Results

This section provides a detailed overview of the data analysis process. First, a quantitative analysis was conducted, and statistical approaches were applied. Then, a qualitative analysis was conducted.

### 4.1. Quantitative Findings

As mentioned earlier, the quantitative analysis involved descriptive and inferential approaches; respondent demographics were first calculated. Regarding their demographics, it is found that 88.3% of study respondents resided in rural areas, 6.1% lived in urbanized areas, and 5.6% marked “Nomadic” as their living location. Concerning social status, 60.2% of respondents indicated that they were single, 32.9% were married, and 6.9% were widowed. Finally, 42.2% of respondents revealed that they were undergraduate-level students, 19.4% were graduate-level students, 18.3% were postgraduate level, 13.0% were working toward a diploma/certification, and 6.6% were pursuing their doctorate. Table 3 presents the respondent demographics.

**Table 3.** Respondent demographics.

| Variables         | Constructs            | N   | %    |
|-------------------|-----------------------|-----|------|
| Living Location   | Rural                 | 333 | 88.3 |
|                   | Urban                 | 23  | 6.1  |
|                   | Nomadic               | 21  | 5.6  |
| Social Status     | Single                | 227 | 60.2 |
|                   | Married               | 124 | 32.9 |
|                   | Widowed               | 26  | 6.9  |
| Educational Level | Undergraduate         | 160 | 42.4 |
|                   | Graduate              | 73  | 19.4 |
|                   | Postgraduate          | 69  | 18.3 |
|                   | Doctorate             | 25  | 6.6  |
|                   | Diploma/Certification | 49  | 13.0 |

#### 4.1.1. Inner Model Testing

As this research involves Structural Equation Modelling, the first step included testing the measurement model. This step included testing the validity and reliability of the measurement model [81]. First, the convergent validity of the model was assessed to

examine the internal consistency among the variables. It was found that most of the factor loading values surpassed the threshold value of 0.5 [82]. Furthermore, the Average Variance Extracted (AVE) values surpassed the cutoff value of 0.5 (artificial intelligence 0.607, customer experience 0.548, customer satisfaction 0.502, and e-commerce operations optimization 0.586).

Furthermore, the construct reliability was assessed by determining Cronbach Alpha and Composite Reliability values. Cronbach Alpha values also surpassed the cutoff value of 0.7 (artificial intelligence 0.859, customer experience 0.701, customer satisfaction 0.753, and e-commerce operations optimization 0.777). Also, the Composite Reliability values surpassed the cutoff value of 0.7 (artificial intelligence 0.868, customer experience 0.730, customer satisfaction 0.755, and e-commerce operations optimization 0.785). Table 4 represents the results of the convergent validity assessment.

**Table 4.** Convergent validity testing.

| Variables                          | Items | Loadings | Ave   | CA    | CR    |
|------------------------------------|-------|----------|-------|-------|-------|
| Artificial Intelligence            | AI1   | 0.841    | 0.607 | 0.859 | 0.868 |
|                                    | AI2   | 0.908    |       |       |       |
|                                    | AI3   | 0.786    |       |       |       |
|                                    | AI4   | 0.813    |       |       |       |
| Customer Experience                | CE1   | −0.015   | 0.548 | 0.701 | 0.730 |
|                                    | CE2   | 0.746    |       |       |       |
|                                    | CE3   | 0.854    |       |       |       |
|                                    | CE4   | 0.596    |       |       |       |
| Customer Satisfaction              | CS1   | 0.081    | 0.502 | 0.753 | 0.755 |
|                                    | CS2   | 0.845    |       |       |       |
|                                    | CS3   | 0.602    |       |       |       |
|                                    | CS4   | 0.691    |       |       |       |
| E-commerce Operations Optimization | EO1   | −0.131   | 0.586 | 0.777 | 0.785 |
|                                    | EO2   | 0.789    |       |       |       |
|                                    | EO3   | 0.819    |       |       |       |
|                                    | EO4   | 0.823    |       |       |       |

As some of the loading values of the factor loadings are below 0.5, the goodness of fit was tested after removing the relevant items. Notably, this helped to examine the extent to which the observed values fit well with the expected values [83]. Thus, the goodness of fit in the current study indicates (see Table 5) that the Standardized Root Mean Square (SRMR) value is 0.004 ( $<0.08$ ), the Non-Fit Index value is 0.942 (b/w 0–1), and the Tucker and Lewis value remains at 1.836 ( $>0.90$ ). The chi-square value is 1.11874 ( $<3.0$ ). Overall, these results indicate a good fit for the structural model [27].

**Table 5.** Goodness of fit.

|            | Obtained Values |
|------------|-----------------|
| SRMR       | 0.004           |
| TLI        | 1.836           |
| Chi-square | 1.11874         |
| NFI        | 0.942           |

Furthermore, discriminant validity was tested by employing the two-step approach suggested by [59]. First, using the Fornell-Larcker criterion indicated (see Table 6) that the constructs do not correlate with each other. In addition, the Heterotrait–Monotrait ratio scale indicated (see Table 7) that the HTMT values also remained below the cutoff value of 0.85 [84], suggesting that discriminant validity is affirmed.

**Table 6.** Fornell–Larcker criterion.

|                                    | Artificial Intelligence | Customer Experiences | Customer Satisfaction | Operations Optimization |
|------------------------------------|-------------------------|----------------------|-----------------------|-------------------------|
| Artificial Intelligence            |                         |                      |                       |                         |
| Customer Experiences               | 0.572                   |                      |                       |                         |
| Customer Satisfaction              | 0.307                   | 0.368                |                       |                         |
| E-commerce Operations Optimization | 0.343                   | 0.328                | 0.45                  |                         |

**Table 7.** Heterotrait—Monotrait.

|                                                              | HTMT  |
|--------------------------------------------------------------|-------|
| Artificial intelligence → Customer Experiences               | 0.572 |
| Artificial Intelligence → Customer Satisfaction              | 0.107 |
| Customer Satisfaction → Customer Experiences                 | 0.368 |
| E-commerce Operations Optimization → Artificial Intelligence | 0.343 |
| E-commerce Operations Optimization → Customer Experiences    | 0.228 |
| E-commerce Operations Optimization → Customer Satisfaction   | 0.450 |

#### 4.1.2. Outer Model Testing

Finally, the outer model is tested to examine the proposed relationships between study variables. However, first, the independent variable's predictive power and effect size are tested using the Coefficient of Determination  $R^2$  and Effect Size  $f^2$ , respectively. Table 8 represents the Coefficient of Determination  $R^2$  and Effect Size  $f^2$  results. It is found that the independent variable shows the predictive power on customer experiences is 24.7% (Moderate), customer satisfaction is 34.0% (Moderate), and operations optimization is 66.3% (Strong). In addition, the effect size on e-commerce operations optimization is 0.831 (large), customer satisfaction is 0.515, and customer experience is 0.327 (large). Table 7 represents the Coefficient of Determination  $R^2$  and Effect Size  $f^2$  results [85].

**Table 8.** Coefficient of Determination  $R^2$  and Effect Size  $f^2$ .

| Variables                          | R-Square | f-Square |
|------------------------------------|----------|----------|
| Customer Experiences               | 0.247    | 0.327    |
| Customer Satisfaction              | 0.340    | 0.515    |
| E-commerce Operations Optimization | 0.663    | 0.831    |

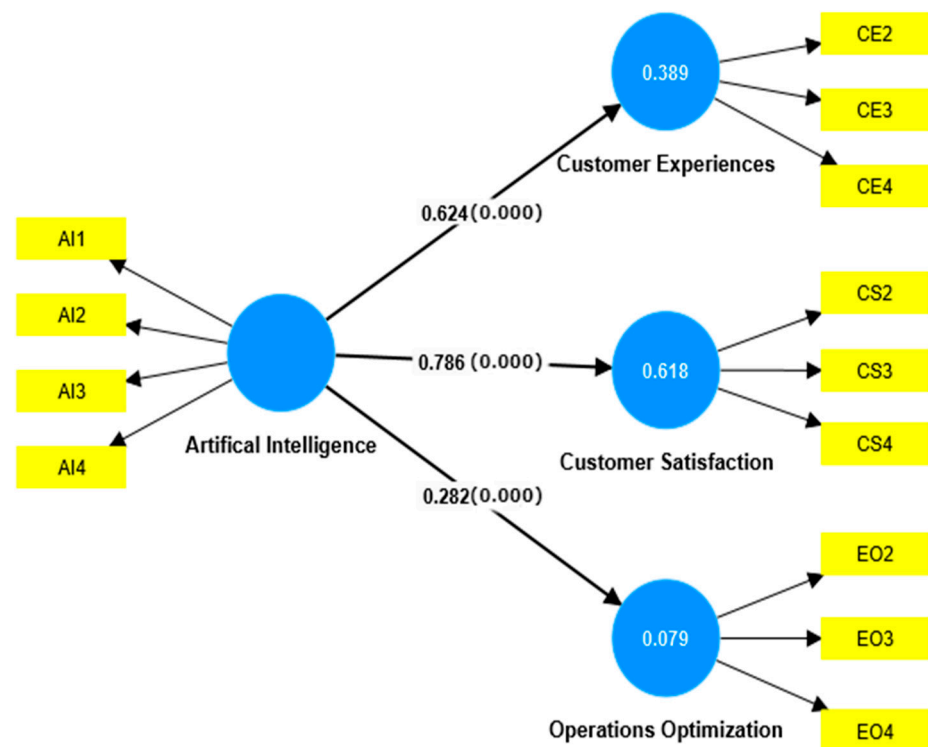
Path analysis was used to examine the proposed study hypotheses [86,87]. Table 9 represents the results of the path analysis, including path coefficients, Means, Standard Deviations,  $t$  statistics, and  $p$  values. First, the proposed effect of AI on customer experiences was tested, indicating a beta coefficient value  $\beta$  0.624 and the significance value of  $<0.000$ , implying that the first hypothesis is accepted. The second hypothesis remains accepted with a beta coefficient value  $\beta$  0.786 and a significance value of  $<0.000$ , suggesting that AI positively affects customer satisfaction. Finally, the third hypothesis was also accepted with a beta coefficient value  $\beta$  0.282 and a significance value of  $<0.000$ . Notably, the path between AI and customer satisfaction remains the strongest, followed by AI and customer experience, and the path between AI and e-commerce operations optimization is the weakest.

These findings show that overall, all of the study hypotheses are significant, showing positive effects of AI on customer experience, customer satisfaction, and e-commerce operations optimization. Figure 1 graphically shows the results of path analysis, where items with low loadings were retained because they represent the core concepts of the latent variables that are difficult to accurately measure using only other items. Although the loadings are low, retaining them does not significantly weaken the statistical power of the

overall model and contributes to a complete representation of the variables studied. This clarification also ensures that readers understand that the model reflects the conceptual reality of the study and that the results are based on all significant indicators, even if some loadings are low [88].

**Table 9.** Hypothesis testing.

| Hypotheses                                                    | $\beta$ | (M)   | (STDEV) | T Statistics | p Values |
|---------------------------------------------------------------|---------|-------|---------|--------------|----------|
| Artificial Intelligence -> Customer Experience                | 0.624   | 0.337 | 0.041   | 10.372       | 0.000    |
| Artificial Intelligence -> Customer Satisfaction              | 0.786   | 0.394 | 0.105   | 7.469        | 0.000    |
| Artificial intelligence -> E-commerce Operations Optimization | 0.282   | 0.401 | 0.372   | 6.3763       | 0.000    |



**Figure 1.** Final structural model.

#### 4.2. Qualitative Analysis

The qualitative analysis involved a thematic analysis of the gathered data suggested by [89,90]. In this regard, the analysis followed the steps suggested by [59]. to ensure the clarity and consistency of results. These steps involved data familiarization and reading, transcription, initial code generation process, identifying themes, re-evaluating the themes, defining themes, and reporting results. Table 10 represents the root questions and themes generated from the gathered data [91].

##### Q1: How Do You Consider AI Implementation in the Saudi E-Commerce Sector?

The first question explores the participants' perceptions of AI implementation in the Saudi e-commerce sector. The relevant question further generated two themes, including perceptions about AI implementation in e-commerce and opportunities for implementing AI in e-commerce.

**Table 10.** Qualitative questions, themes, and codes.

| Root Questions                                                                                                | Themes                                                      | Codes |
|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------|
| How do you consider AI implementation in the Saudi e-commerce sector?                                         | Perceptions about AI implementation in e-commerce           | E1    |
|                                                                                                               | Opportunities to implement AI in e-commerce                 | E2    |
| In your opinion, how does AI affect customers in Saudi online retail?                                         | Impact of AI on customers                                   | E1    |
|                                                                                                               | AI-driven personalization and customer engagement           | E2    |
| How would you describe the current framework of AI in Saudi e-commerce?                                       | Structure and components of AI in Saudi e-commerce          | E1    |
|                                                                                                               | Current capabilities                                        | E2    |
|                                                                                                               | Challenges and limitations for AI implementation            | E3    |
| What recommendations would you suggest for improving the implementation of AI in the Saudi e-commerce sector? | Strategies for enhancing AI performance in Saudi e-commerce | E1    |
|                                                                                                               | Best practices and innovation for AI implementation         | E2    |

### Theme 1: Perceptions About AI Implementation in E-Commerce

The first theme from the gathered responses indicated an overall positive perception of AI in Saudi E-commerce implementation. According to [92], AI plays a crucial role in retail and e-commerce, as it efficiently predicts customer demands, automates store operations, improves customer engagement and experiences, and optimizes pricing [55]. Hence, current study participants indicated the role and effect of AsI as a positive addition due to facilities like automation and personalization. For example, **participant 3 argued that** “As an online retail store owner, I see AI implementation as a game-changer for the e-commerce sector in Saudi Arabia. It has upgraded many of our processes, from inventory management to customer services, helping my company work more efficiently and effectively to meet customer demands.” In Line with Participant 3, Participant 7 Further Opined That

“AI has brought many opportunities for us to improve our retail services. We can analyze customer data and personalize marketing efforts. However, despite it being a positive addition, it still has many improvements. I am optimistic about its current integration and government support for AI implementation in the e-commerce industry”.

### Theme 2: Opportunities in Implementing AI in E-Commerce

The next theme from the first root question indicated that all study participants agreed that AI implementation offers many opportunities. Automating the redundant tasks, predictive analytics, and resource allocation remained the most prevalent responses. As noted by [55] while AI has existed since the 1950s, its popularity has surged in recent years due to its ability to create business value [55]. It helps retailers predict future demand, manage promotions, and improve the delivery of goods and services to customers. Consistent with the relevant argumentation, **participant 1 argued that** “One of the biggest opportunities AI offers is predicting customer behavior and trends. This helps us stock the right products at the right time, reducing waste and increasing sales. AI has also automated redundant tasks, freeing our employees to focus more on strategic activities”. **participant 2** “One of the most significant opportunities offered by AI is its ability to predict customer behavior and trends, which helps companies provide the right products at the right time, reducing waste and boosting sales” **participant 3** “Artificial intelligence applications have contributed to the automation of routine and redundant tasks, allowing employees to focus on strategic, value-added activities. This transformation represents a key element in increasing operational efficiency and enhancing companies’ competitiveness in the Saudi e-commerce market.” **Participant 4 Further Added That** “AI provides us with advanced analytics and previously unavailable insights. This enables us to make more informed decisions about our marketing

strategies, pricing models, and customer engagement techniques. The potential growth and improved customer satisfaction through AI is immense.”

## **Q2: In Your Opinion, How Does AI Affect Customers in Saudi Online Retail?**

The second root question analyzed the participants’ responses regarding the effect of AI on customers in Saudi online retail. The collected data showed two main themes from the responses: the impact of AI on customers and AI-driven personalization and customer engagement.

### **Theme 1: Impact of AI on Customers**

The first theme generated from the collected responses indicated that study participants consider AI to affect customers positively. As noted by [58], AI is revolutionizing online retail structures as automated retail stores powered by the relevant technology represent the next significant advancements in physical retail, providing customers with a fully automated shopping experience. Thus, **according to participant 3**, “AI significantly improves customer experience by providing personalized recommendations and faster customer services. For example, chatbots can handle questions round-the-clock, ensuring customers get immediate responses and support, promoting over satisfaction and loyalty.” **Participant 6 Further Argued That** “From the customers’ service perspective, AI helps predict customer needs and preferences, leading to improved shopping experiences. This not only enhances customer satisfaction but also improves their experience. Their shopping experiences are mediated by AI-enhanced systems, making them feel valued and understood”.

### **Theme 2: AI-Driven Personalization and Customer Engagement**

The second theme generated from the second root question reflects the participants’ opinions about personalization and customer engagement enabled by AI. Participant 7 argued that “AI-driven personalization allows us to offer customized product suggestions based on individual browsing and purchase history. This makes customers feel like the shopping experience is designed specifically for them, significantly improving their engagement and satisfaction.”

As noted by [88], the combination of AI and predictive analytics gives retailers unique insights into customer preferences and market trends. AI uses machine learning algorithms, computer vision, and natural language. However, retailers can derive practical insights from large datasets, helping them predict trends, manage inventory, and enhance the customer decision-making process. According to participant 2, “AI enables us to send personalized marketing messages and promotions by analyzing customer data. This targeted approach increased customer engagement and drove higher conversion rates, as customers are more likely to respond to offers that resonate with their specific preferences and behaviors.”

## **Q3: How Would You Describe the Current Framework of AI in Saudi E-Commerce?**

The third root question focuses on analyzing the participants’ perceptions regarding the current framework of AI in the Saudi e-commerce sector. The relevant question further led to the generalization of three primary themes, including the structure and components of AI in Saudi e-commerce, current capabilities, and challenges and limitations for AI implementation.

### **Theme 1: Structure and Components of AI in Saudi E-Commerce**

According to [93], the advancement of AI provides retailers with several opportunities. It mainly involves systems and programs that imitate human intelligence through technologies, including machine learning, natural language processing, image recognition, and data

mining. Implementing these systems in online retail services supports the overall system and indicates that it is aligned with mandatory requirements consistent with technology-enhanced services. Hence, the first theme of the third root question reflected several key components of AI in Saudi e-commerce sectors. These components are indicated as having positive, constructive effects on the overall performance of the relevant sector. **As Participant 7 Argued** “The current framework of AI in Saudi e-commerce consists of many key components, including advanced analytics tools, machine learning algorithms, and strong data management systems. These elements work together to improve the performance and overall customer satisfaction.” **Participant 4 Further Opined That** “AI’s structure in the Saudi e-commerce sector incorporates several technologies, such as chatbots for customer service and recommendation engines for personalized shopping experiences. This integration creates a comprehensive system that supports both customers and business.”

### Theme 2: Current Capabilities

The second theme generated from the collected data revealed the capabilities of AI in Saudi e-commerce systems. As noted by [77] these systems have capabilities like automation, predictive analytics, machine learning, natural language processing, and others that imply that they are consistent with strong enhanced technology capabilities [76]. These responses highlighted predictive analytics, automation, and inventory management as robust capabilities of AI technology. **Participant 1 revealed that** “AI in Saudi e-commerce currently enables businesses to analyze customer data in real-time, allowing them to understand buying behaviors and preferences. This capability helps retailers design their offerings and marketing strategies effectively.”

According to Participant 5 “One of the significant capabilities of AI right now is its ability to automate various processes, from inventory management to customer service. This automation saves time and gives us spare time to design, implement, and monitor effective strategies. These steps further help improve overall shopping experiences for customers.”

### Theme 3: Challenges and Limitations for AI Implementation

According to Kozlovskaja [85] lacking specified infrastructure and professionals to analyze, monitor, and manage AI-enhanced systems is a major challenge for implementing relevant technology in different sectors [94]. Talking specifically about AI implementation in e-commerce, these challenges need robust designs and implementations of effective strategies to counteract and improve overall performance. Therefore, the third theme of the third root questions explores the participants’ opinions about the challenges and limitations of effective AI implementation in the Saudi e-commerce sector. The study participants indicated difficulties in the current infrastructure for incorporating AI and lack of skilled workforce as barriers to effective AI implementation.

As Participant 1 Argued “There must be many challenges. However, one of the major challenges, I think, is the lack of skilled professionals who can manage and develop AI technologies specializing in e-commerce operations. This skill gap hinders the adoption of AI and requires careful consideration”. According to Participant 2 “According to my experience and opinion, the integration of AI systems with the existing infrastructure is due to the lack of planning and implementation. Many retailers face difficulties seamlessly incorporating AI into their daily operations, further slowing the overall implementation process. These difficulties are challenging for an overall implementation of AI in the e-commerce sector and need to be overcome”.

### Q4: What Recommendations Would You Suggest for Improving the Implementation of AI in the Saudi E-Commerce Sector?

Finally, the last root question examines the participants’ responses regarding their suggestion to improve AI implementation in Saudi e-commerce. The responses to the

relevant question further led to the generalization of two themes, including strategies for enhancing AI performance in Saudi e-commerce and best practices and innovation for AI implementation.

### **Theme 1: Strategies for Enhancing AI Performance in Saudi E-Commerce**

The study participants proposed primary strategies to counteract the challenges and improve AI performance and implementation in the Saudi e-commerce sector. The existing literature [1,43,49,51] also emphasizes that improving technological integration in the e-commerce sector offers many benefits as it has many practical uses. This technology increases business efficiency, enables retailers to add more stock, and enhances customer service. AI also helps write market collaterals, helps customers when human service is not available, and identifies suspicious financial activities. Consequently, improving the infrastructure and implementing new strategies and best practices for AI implementation in e-commerce sectors is of greater significance. According to Participant 4 “One of the crucial strategies to improve AI implementation and performance is to address the skill gap in the industry. We need to invest in specialized training programs, and courses focused on AI and its applications in e-commerce. Partnering with educational institutions to create certification programs can help develop a workforce proficient in AI technologies. Also, offering incentives for continuous professional development in AI can motivate current employees to improve their skills, ensuring that we have the expertise needed to manage and develop advanced AI systems”. Participant 6 Further Added That “To improve the implementation of AI systems with existing infrastructure, I suggest a phased approach to AI adoption. This includes starting with small-scale pilot projects that allowed retailers to test and refine AI technologies in a controlled environment before full-scale implementation. Developing a clear road map that includes thorough planning, stakeholder involvement, and step-by-step integration of their help addresses the challenges of incorporating AI into daily e-commerce operations. Also, creating a supportive network or a consortium of retailers can facilitate knowledge sharing and provide practical solutions to common implementation issues.”

### **Theme 2: Best Practices and Innovation for AI Implementation**

Füller et al. [95] argued that implementing AI in e-commerce necessitates that the technology is effectively designed and implemented to fulfill customer needs, improving their experiences [85]. This approach helps to optimize AI systems for improved efficiency and accuracy, enhancing customer satisfaction. In addition, it also provides a framework for addressing common challenges, such as data privacy and security, ensuring that AI is deployed ethically and complies with regulations. Therefore, these practices promote innovation, drive operational efficiency, and help retailers stay competitive in an evolving market. The final theme developed from the last root question was based on the participants’ opinions about best practices and innovation for AI implementation. In addition to recommendations for the improved framework, they opined necklaces and innovations that could improve the current AI implementation in the Saudi e-commerce sector. According to Participant 6 “As e-commerce is booming today, it is important to brainstorm strategies and best practices that can improve AI implementation for favourable results. In this regard, one best practice for AI implementation is to adopt A customer-centric approach. This means continuously collecting and analyzing customer feedback to refine and modify AI systems and ensure they effectively meet user needs. Using AI to personalize customer interactions can significantly improve user experience and satisfaction. Innovating by using AI-driven chatbots and recommendation systems that adapt based on real-time data can also create a more engaging shopping experience”. According to Participant 7 “One best practice is to stay ahead of technological advancements by investing in research and devel-

opment. Retailers should establish a dedicated AI innovation team to explore emerging AI technologies and their potential applications in e-commerce. Collaborating with tech startups and participating in AI-focused industry conferences can provide fresh insights and innovative solutions. Besides, ensuring strong data security and privacy measures will build customer trust and compliance with regulation, encouraging a reliable environment for AI deployment.

## 5. Discussion

This research analyses AI in the Saudi e-commerce sector from the perspectives of both customers and retail owners. Theoretically supported by social and technological theory, this research highlights how the design, implementation, and performance of e-commerce retail in Saudi Arabia is improved considering both social and technical improvements offered by AI implementation. On the other hand, these results also remain consistent with the proposition by [37], as the gathered data identifies some challenges hindering the implementation of AI in the Saudi e-commerce sector. However, technological advancements have also been found to influence Saudi e-commerce retailers and suggest improvements in the existing systems to acquire maximum benefits from technological integration. Overall, this study indicates a common agreement by both customers and retail owners about the usefulness of AI implementation in e-commerce, consistent with the existing literature on e-commerce and its prospects [50,78,79,96]. Considering the obtained data, the study respondents agreed that AI in online purchases is a positive technology addition today, adding more value to daily shopping experiences. According to the respondents, AI in online shopping and retail is positively affecting product decision-making, as it is facilitated by AI-enhanced features like chatbots and virtual assistants that are intuitive for buying online products. As noted by [91]. AIS emerged as a popular topic in the e-commerce world, as it aims to improve existing human–computer experiences and interactions [91]. AI systems act quickly and offer practical solutions based on the data gathered from different resources. The decisions made by AI are effective, as they depend on the real world or comparable data. This entire framework functions according to established protocols designed for specific tasks. Notably, the quantitative part of this research is guided by three study hypotheses.

The gathered data supports the first study hypothesis, “**H1. AI has a positive effect on female Saudi customers’ experience in the Saudi Arabian e-commerce industry**”. The study respondents agreed that they were provided with correct solutions by the AI chatbots used by the service providers, as their queries were adequately addressed after talking to the digital systems. According to the respondents, they felt engaged when AI Chatbots collaboratively addressed the required products, and the visual cues on the website interface were convenient. Furthermore, qualitative participants indicated that AI enhanced their customer experience by providing a personalized recommendation system and a quicker customer support service. Another important aspect is predictive analytics, which predicts customers’ needs and preferences, leading to improved shopping experiences, increased customer loyalty, satisfaction, and positive feedback. Another study [97] also witnessed how AI is used and implemented by the online retail sector in India to improve customer experience and enhance profitability. According to the researchers, AI provides information about customer preferences and online behaviors. It relies on their search history, and provides tailored suggestions to help them consider and choose products and services that may match their requirements. Customers and retailers equally consider AI systems to provide a pathway of improvement and positive experiences for both customers and retailers, leading to increased reliance and satisfaction on digital technologies.

Regarding the second study hypothesis, “**H2. AI has a positive effect on female Saudi customers satisfaction in the Saudi Arabian e-commerce industry**”, the results remained supportive. According to the study respondents, they have had positive experiences when interacting with AI-enabled online retailers, and they would readily recommend an online site that uses AI. The study respondents further stated that their shopping experience with an AI-enabled online retailer website has fulfilled their expectations, and they are generally satisfied with their purchases from the relevant platforms. Consistent with the quantitative data, qualitative responses also indicated an overall positive impact of AI on customer satisfaction in e-commerce. The study participants emphasized AI-driven personalization, providing customized product suggestions based on individual browsing history. This made customers feel like the shopping experience was designed according to their needs, further improving their satisfaction. In addition, AI’s personalized marketing messages and advertisements were also acknowledged, implying their consistency with the customer’s needs, leading them to attract and find products and services, resulting in customer engagement and satisfaction. In line with these findings, a study by Nagy et al. [98] also examined AI implementation, acceptance, and use among Hungarian online customers by using the technology acceptance model [97]. Data collected from quantitative surveys indicated that Hungarian customers consider AI implementation a significant contribution to the grant framework of e-commerce. The study respondents indicated that the wider acceptance and use of AI systems in the Hungarian e-commerce sector is due to the perceived trust, usability, and user satisfaction linked with technology usage. These results remain consistent with the responses of Saudi e-commerce customers and retailers, who equally consider AI to be an important aspect of modern online retail systems.

Finally, the third hypothesis is “**H3. AI positively optimizes e-commerce operations in the Saudi Arabian e-commerce industry**”. According to the study respondents, they believe that AI tools significantly improve the efficiency of online retail shop operations, as they help inventory management functions in these businesses. The respondents further agreed that the use of AI contributes to more rapid order fulfillment in online stores, leading them to believe that today’s online purchase-related decision-making is largely simplified by relevant technological advancement. The qualitative participants also indicated a similar opinion about AI implementation in their online stores. They generally indicated that AI has significantly enhanced their overall system by improving the operational speed, efficiency, and quality of customer service. In addition, improvements in inventory management systems and order processing are important features that AI offers in the e-commerce sector. Accordingly, Pallathadka et al. [96] argue that technology is improving in tandem with scientific advancements, leading to substantial changes in how individuals and businesses operate. Referring particularly to e-commerce and AI, relevant technology has become a critical tool for driving market growth and improving operational efficiency. In this regard, several AI-enabled technologies, such as recommendation engines, AI assistants, optimal pricing strategies, order processing systems, stock management, and many others, are contributing effectively to improving the functions of the e-commerce industry.

However, despite the robust advancements offered by AI technology in the Saudi e-commerce sector, the study participants also indicated some grassroots-level challenges that need to be addressed. These challenges are called skill gaps, and they include the need for adequate and often effective AI implementation. To counter these issues, the study respondents suggested some strategies and recommendations to improve the AI framework in the Saudi context, implying its significance and prospective role for future e-commerce operations. As also emphasized by [99] addressing the challenges linked with AI in e-commerce is important for maximizing its potential benefits and ensuring sustainable growth in the sector. These challenges potentially hinder effective AI implementation and

diminish consumer trust. Thus, businesses can enhance operational efficiency, promote innovation, and improve customer experience by countering these obstacles. Successfully overcoming these challenges not only leads to a stronger AI system, but also positions companies to better adapt to market demands and stay competitive in an increasingly digital landscape [5,72]. In this study, sociotechnical theory was employed to understand artificial intelligence and e-commerce in the Saudi context. The results show that active participation in the use of digital systems contributed to modifying and improving the procedures and services provided, reflecting a process of joint improvement between the social and technical aspects. For example, feedback provided by customers and business owners helped organizations develop more user-friendly interfaces and tailor services to their actual needs. This provided ongoing support to employees and enabled channels of communication with customers, enabling them to achieve higher levels of efficiency and respond quickly to market demands. In addition, the data show that continuous modifications and improvements to digital systems were the result of interactions between users and technologies (co-optimization), while the interconnectedness of technical and social systems contributed to enhanced operational efficiency and higher levels of customer satisfaction. This link between theory and findings demonstrates that organizational and digital effectiveness is not achieved through technology alone, but rather through the integration of technical and social dimensions, which reflects the essence of sociotechnical theory and strengthens the explanatory power of this research.

## 6. Recommendations

Based on the study results, this research provides practical recommendations to the e-commerce stakeholders, government officials, and other individuals involved in the relevant industry. First, there is a need to address the scale gap in the industry by investing in specialized training programs and courses focused on AI and its applications. Teaming up with educational institutions and creating certification programs can help to develop a workforce skilled in AI technologies. It is also suggested that incentives for continuous professional development in AI be offered to motivate grant workers to enhance their skills and develop advanced AI systems. Another important recommendation is the development of a phased approach. Starting with small-scale pilot projects helps retailers test and refine AI technologies before large-scale implementation. Developing a clear road map can help handle the challenges linked with AI implementation. The third important consideration is the adoption of a customer-centric approach. It is true that constantly collecting and analyzing customers' feedback can help improve AI systems to meet users' needs actively. It is also important to stay ahead of technological advancement by spending on research and development. Establishing dedicated AI innovation teams can provide fresh insights and innovative solutions. Teaming up with technology enterprises and partaking in a focused industry gathering study can further improve innovation. Data privacy is also important to build customer trust and compliance with regulations. Implementing robust data protection strategies can create a reliable environment for AI implementation, encouraging the more widespread adoption and acceptance of AI. Finally, this research highlights the significance of enhancing existing AI frameworks by incorporating advanced analytical tools, machine learning algorithms, and a strong data management system. These components can improve performance, operational efficiency, and customer satisfaction in Saudi e-commerce.

## 7. Conclusions

AI technologies have revolutionized businesses, providing sources for better customer insights, personalized shopping experiences, and streamlined operations. The results of

this research confirm the vital role that artificial intelligence plays in enhancing the efficiency of e-commerce operations in Saudi Arabia. This study demonstrates that the use of AI tools and technologies, such as big data analytics, demand forecasting systems, and intelligent personalization of marketing experiences, effectively contribute to improving business performance, increasing customer satisfaction, and maximizing sales. This analysis demonstrates that a company's ability to integrate technical aspects with the social and organizational dimensions of business operations is a critical factor in the successful adoption of these technologies. This aligns with the concepts of socio-technical theory, which focuses on the interconnectedness of systems and joint optimization. This study also demonstrates that Saudi companies are adopting AI not only to improve internal operations, but also to provide personalized services that meet diverse customer needs, enhancing their competitiveness in a rapidly evolving digital market environment. Through qualitative and quantitative analysis, it was found that the data collected and analyzed by these systems provides strategic insights that enable companies to make more accurate and effective decisions, whether related to inventory, marketing, or customer relationship management. However, this research indicates that fully leveraging AI's potential requires improving digital infrastructure, providing ongoing training for human resources, strengthening data management systems, and fostering a culture of innovation within organizations. Investing in these areas enables companies to maximize AI's potential, ensuring their continued competitiveness and sustainable growth.

## 8. Study Limitations and Future Research

Although this research fills an important gap in the existing literature, it has some limitations that must be considered in future studies. First, this research is based on university-level female students in Saudi Arabia, questioning the generalizability of results to other sectors and regions. Future researchers can delimit this scope by focusing their studies on more diverse samples based on demographic, gender, and institutional aspects. The second limitation involves our employment of the convenience sampling technique for qualitative data gathering. Future researchers can employ other different sampling strategies that may not involve researchers' own biases in order to overcome this limitation. The study results indicate the trends and preferences of female undergraduate students at two public higher-education institutions, highlighting the characteristics of this particular group. While the results reflect the behaviors and attitudes of the target sample, it should be noted that they do not necessarily represent all clients or students in the Kingdom of Saudi Arabia. Therefore, these findings can be used to develop educational or marketing strategies or programs specifically targeted at female students at similar higher-education institutions, keeping in mind that transferring the findings to other contexts may require additional research. Finally, applying the qualitative approach also involves some primary limitations, mainly regarding the generalizability of results to a relatively smaller sample size. Future studies can also delimit this concern by adding and employing other approaches, as well as enhancing their studies' scope.

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